then go hunting for an engine. It has to be a close association between engine manufacturers and aircraft designers. And if you do it independently, the mismatch will always be there," says a key defence official.

For the Tejas, the Aeronautical Development Agency (ADA) had to settle for GE F404 engines (90kN), reducing the aircraft's payload and range. Similarly, the GE F414 engine with 95-98kN power thrust, planned for Tejas Mk2, is sub-optimal for the IAF's specifications in the medium-weight fighter class of 15-18-tonne range—to enable a 5-tonne combat payload and a range of 600 km on internal fuel, says a key IAF official, who was in the LCA's development team.

The ADA, also the developer of AMCA, is looking for 110 kN thrust—which is not yet available globally—to meet the requirements of supercruise features in its upgraded version, the AMCA Mk2. It is expected to go into production from 2035 onwards. The LCA Mk2 is supposed to replace the IAF's 16 fighter squadrons, including three of Mirage 2000, five of the MiG-29s, six of the Jaguar strike aircraft and the two remaining MiG-21 Bison squadrons. With this calculation, IAF is looking at a combat strength of about 40 squadrons by 2040.

After Kaveri's failure, the LCA makers had no option but to turn to foreign engines. In 2005, GTRE sought a foreign hand for the co-development with an international engine-house for a 90kN class engine. But the plan never worked. In 2012, France offered complete transfer of technology (ToT) for a redesign and fixing of the Kaveri engine to fit into the LCA Tejas. But the deal could not go through due to its high price.

T. Mohan Rao, the former head of GTRE, who was closely involved in the Kaveri project, says, "We had reached up to achieving 70 per cent of technology, but for the remaining, we required handholding by a foreign player." Though there are nearly 40 aircraft manufacturers in the world, only three countries—the US, Russia and France—have the technology to completely design, develop and produce military aircraft engines, he adds. The Kaveri programme has been shelved except for the possibility of using the engine for the Ghatak armed drone.

Interestingly, the US has revived its earlier proposal for collaboration on jet engine technology under the India-US Defence Trade and Technology Initiative, which was suspended in October 2019 owing to American reluctance to share technology. Tata is already in partnership with GE to produce parts for a civilian jet engine.

Similarly, France's Safran has already partnered with Hindustan Aeronautics Ltd (HAL) to make the indigenous Shakti engine for helicopters. Days after the Rafale deal in 2015, Safran pitched a technology transfer of engines as part of the offset clause, but it didn't take off. British firm Rolls-Royce, which has allied with HAL, is now keen on the co-development of a 110 kN engine for the AMCA. Kishore Jayaraman, president, India and South Asia, Rolls-Royce, says India is a key growing market for the company. "We continue to work closely with the MoD (ministry of defence), DRDO, HAL and the private sector to explore opportunities to create in India products and solutions for the Indian defence sector," he adds.

Safran offers complete ToT for design, development and manufacture at close to $5 billion. It will provide everything for engine manufacturing, from knowhow to certification. Satish Kirtikar, former vice president of Safran Aircraft Engines, claims that India can achieve independence in aircraft engine technology only if it is able to purchase the "knowhow and know-why" from a foreign OEM. But this ToT will come with a heavy price tag. Kirtikar says that private players in India will have a hard time entering the aerospace and defence arenas, justifying huge investments to their boards, with the returns on investments taking at least seven to 10 years to yield any sizeable returns. "Besides, these businesses will require work guarantees over an extended period of time to set up their facilities," he explains.

India's military planners have to make some hard choices: between *aatmanirbharta* and dependence. Only then will its ambitions take flight. ■



INDIA'S JET ENGINE JOURNEY

Without its own jet engine, India has to settle for inadequately powered imported engines for its LCA Tejas fleet. The Gen 5 fighter jet programme, AMCA, has also been limping for the past 13 years as it seeks an engine with 110 kN thrust to meet its supercruise requirements

DEC 1986
DRDO's Gas Turbine Research Establishment begins work on the Kaveri engine for the LCA with a projected cost of Rs 382.86 crore

1989
The budget for the 93-month Kaveri engine programme is approved in March. A full-scale development is authorised in April

1996
Test runs of the first complete prototype of Kaveri begin and all five ground-test examples are in test by 1998

2003
Kaveri is sent twice to Russia to undergo high-altitude tests; it fails the tests. A decision is taken to procure the uprated GE F404-IN20 engine for the LCA

2008
Despite having undergone 1,700 hours of tests, the Kaveri is delinked from the Tejas programme

2012
India offers Safran a contract to fix the Kaveri engine. The deal fails due to high costs

2015
Safran again offers the engine development proposal as part of the Rafale deal offset obligations. No decision yet